Polarization Imaging

Introduction

Objective

The main goal is to understand how a simple camera can be transformed into a polarization state measurement system. This lab introduces two different systems:

a simplified polarization imaging set-up that consists of a manually rotating polarizer placed in front of the sensor, a contrast polarization measurement system that uses a Twisted Nematic Liquid Crystal.

Equipment

PC computer

Camera via USB cable

Arcoptix switchable polarization rotator 0-90 (Twisted Nematic Liquid Crystal) + Arcoptix USB LC Driver

Two linear polarizers

Four mounting posts and four post holders

Lighting Device + Polarized Ring

Software

Acquisition via IDS software or Matlab

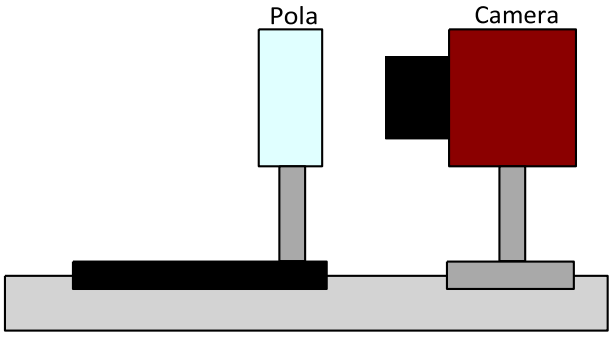
Arcoptix USB LC Software

Matlab

Simplified polarization imaging

Getting started

1. Design the following imaging set-up:



2. Connect camera cables.
3. Start the acquisition software.

4. Select the connected camera.

5. Grab a part of the computer screen and rotate the polarizer to see what happens .give details of what happens ?

Wolff's method

This method is in fact a special case of the Least Mean Square Method...

.Before taking images, Settings of the camera like autoexposure and gain must be set to manual: why ?

1. Snap and store 3 images of a scene made of a part of the computer screen with the followingpolarizer orientations: 0 , 45 and 90 .

2. With Matlab or LabVIEW compute and show the polarization parameters of the scene givenby:

$$\begin{cases} I & = I_0 + I_{90} \\ \tan 2\varphi & = \frac{I-2I_{45}}{I-2I_{90}} \\ \rho & = \frac{\sqrt{(I-2I_{45})^2+(I-2I_{90})^2}}{I} \end{cases}$$

.Describe the images ϕ and ρ according to the regions of interest.

3. Compute a color image that can represent the three parameters on a single image.

Least Mean Square method

1. Snap and store N (N = 8 for instance) images of a scene made of a part of the computer screen with N orientations of the polarizer.

   Questions 2-4 are optional during the lab and can be treated later.

2. Apply the Least Mean Square method to the images to estimate first s_0, s_1 and s_2.

3. Compute and show I, ρ and ϕ.

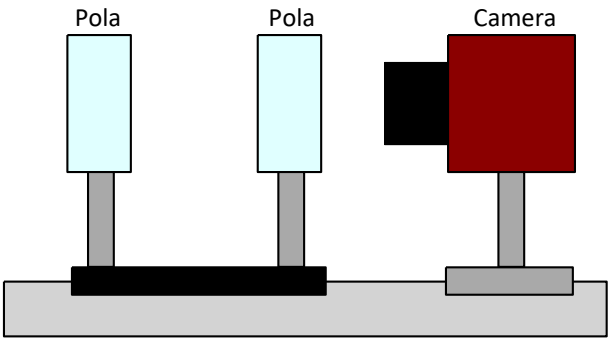
4. Draw for some pixels of interest the light intensity measured by the camera according to theangle of the polarizer to show the sinusoidal relationship .detail the results.

Contrast polarization measurement

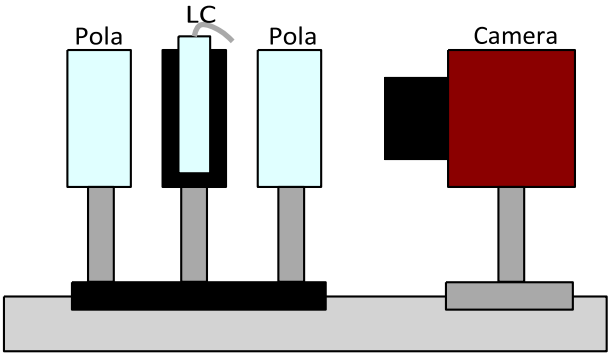
This method consist in taking two different images with two orthogonal directions of a linear polarizer. If an additional polarized lighting according one of these directions is used it enables to estimate the depolarization properties of the object and to remove specular reflections. A liquid crystal rotator is used here to programmaticaly change the orientation of the polarizer.

Getting started

1. Design the following imaging set-up with the two polarizers oriented at 0 and check how light intensity generally decrease (explain why):



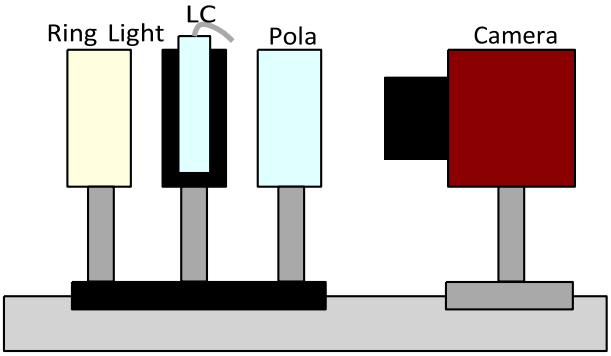
2. Mount the Arcoptix switchable polarization rotator 0-90 , insert it between the 2 polarizers:



3. Connect Acroptix usb LC Driver to the computer.
4. Grab a scene and play with the Arcoptix software the polarizer to see what happens.
   .explain why ?

Diffuse Specular reflection

1. To remove specular reflections an additional polarized lighting source can be used. Check thatthe polarizer used in front of the light is parallel to the polarizer in front of the camera:



2. Snap two different images of an object (your hand for instance, but don't move) and store theseimages I_k and $I_\perp$.
3. Compute and show the following parameters:

   Total light intensity: $I_k + I_\perp$

   Polarization contrast: $I_k - I_\perp$ Polarization contrast ratio: $\frac{I_\parallel - I_\perp}{I_\parallel + I_\perp}$ .Show areas containing specular or diffuse reflection.

4. If you have time and know LabVIEW programming, create a program that automatically provides the previous parameters.

End of Lab Course

Disconnect all Cables.

Uninstall every optical devices very carefully and store everything as it should be Carefully

Store every components in the box.

Appendix

Some useful Matlab functions imread : to read images

double : to return the double precision of an array figure

: to create a figure imagesc : to show images colormap : to

change the colormap colorbar : to show the colorbar

hsv2rgb : to create a RGB image from the HSV values

Laboratory Course n° 2

Infrared Imaging

Objective

The goal of this practical work is to study the application of IR imaging to temperature measurement and to observe the influence of several parameters on the efficiency of these measurements.

Equipment

- PC Computer
- IR Camera FLIR
- Hotplate and Chauvin Arnould Lab equipment
- Industrial Parts and Electronic Circuit

Software

Matlab (icon on the main desktop) with scripts

Documentation

Technical Manual of the camera

Study of the camera

Getting started : get your first image

The camera is connected on the Ethernet port of the computer. Plug the camera too power it, boot the PC with Windows and launch Matlab.

You have to wait that the temperature of the camera is stabilized first obtain infrared images (5 minutes).

To test the if the Ethernet connection between the camera and the PC works well, open the FlirView application. You should observe a first gray image. With low contrast.

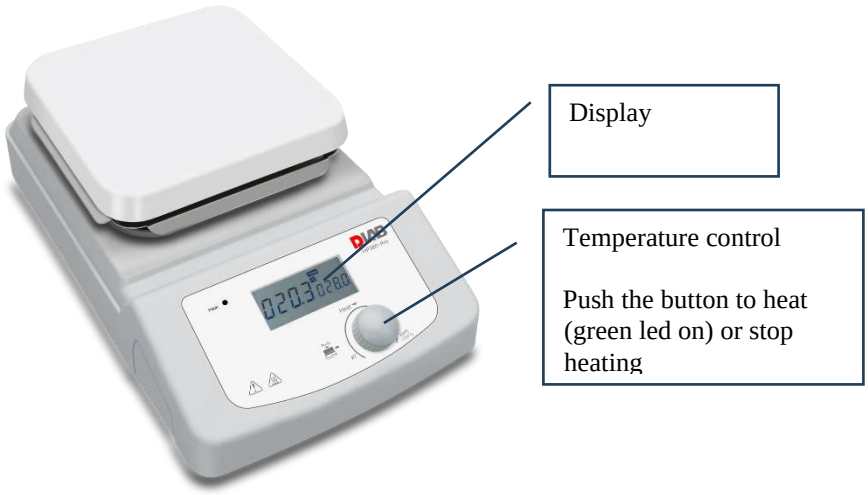
Now you can open the "flir_acquisition.m" script. Identify and comment each part of the script. (Logically no modification should be done; in case please change the name of the file) .

Study how the images are saved, what is the format ? .

Camera calibration

Using the hot plate also equipped with a digital thermometer, the goal of the experiment is to realize the temperature calibration of the camera (ie the link between gray level (digital level) and the temperature.

The screen of the hotplate displays: on the right the desired temperature, on the left the one measured by the probe.



Do not tune the temperature of the plate more than 150 Celcius degrees in order not to damage the camera.

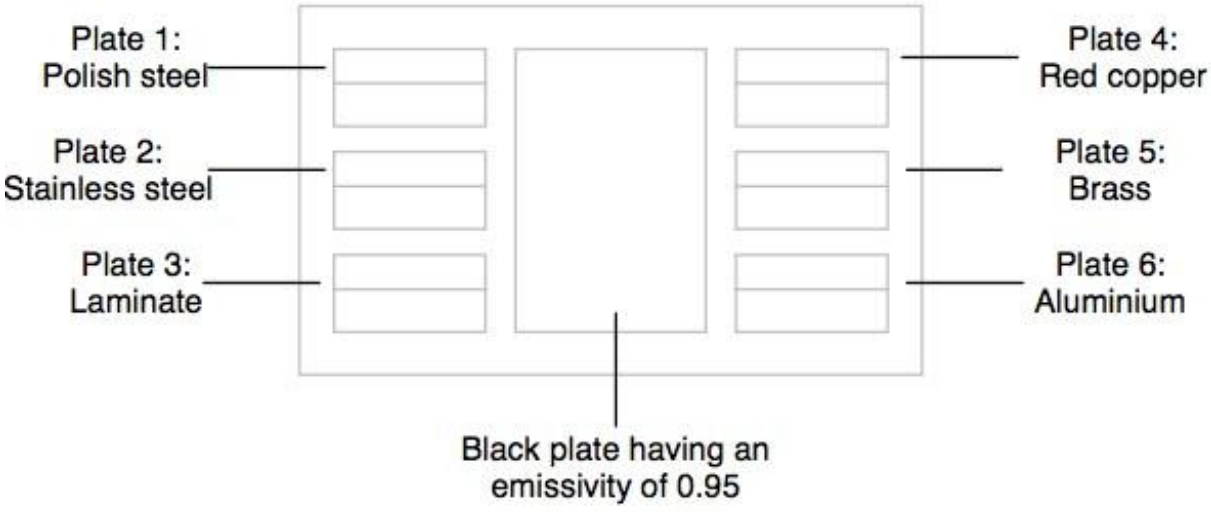
Present onto a 2D graphic the relationship between the real temperature and the digital level (DL).

Give some comments and critics on the calibration process. Do you think that the infrared source is good for this experiment? Why?

Estimation of emissivity

The **C.A 1875 tutorial bench** comprises a hot plate with several targets having different surface conditions and made of different materials, along with test screens that are affixed to the front of the bench using magnets. (See the diagram below).



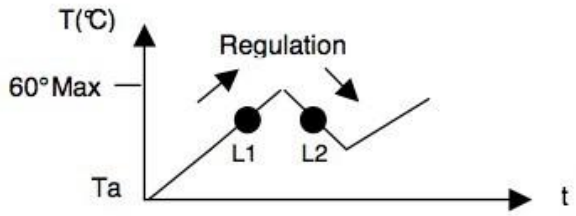


Before strating the bench, take and save an image of the bench with the IR camera.

What do you observe ?

Start the bench.

When the bench is started up, L1 lights until the plate reaches approximately 55°C. When this temperature is reached, L2 lights and the plate cools to approximately 50°C. Heating resumes, with the lighting of L1, and so on. This cycle continues indefinitely for as long as the bench is in operation.



When the temperature of the bench is stabilized, take a second image.

Determine the emissivity of each material of the bench (we assume that the emissivity of the black plate is 0.95).

Transmission and reflection coefficients

Transmission coefficient.

Using the bench presented in the previous section, propose an experimental setup witch can be used to determine the transmission coefficient of a glass plate.

Do the same experiment with black PVC.

Reflection coefficient.

Propose an experimental setup witch can be use to determine the reflection coefficient of a piece of copper.

Laboratory Course n° 3
–
Lens & Lighting

Objective

Taking into account the scene you have to image and features to be detected, you have to choose:

- Lens
- Lighting configuration to implement.

Equipment

- PC Computer
- IDS Digital Camera, USB Cables
- Industrial Lenses
- Extension Rings
- Lighting Devices
- Industrial Parts

Software

IDS software (Uye Cockpit)

Documentation

- User Manual
- Lectures on Machine Vision

Laboratory Course n° 3
–
Lens & Lighting

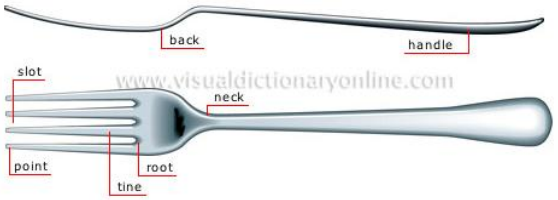
Lens

In order to carry out this step, consult the following documentation:

- Camera Brochure
- Lectures on Machine Vision

You have to observe the fork in order to perform surface inspection with the following specifications:

- Working Distance : 500 mm



Choose the right lens, by using the estimation formula in the course.

Mount the selected lens on the camera and check that the computation is OK:

1. Connect Camera cables.
2. Start Ueye cockbit or demo software.
3. Select the right camera and start Grabbing images.

Ask professor to validate once everything is OK.

Compute the obtained "Spatial Resolution"

Select different settings and study performances w/r to full or partial frame acquisition => Select "Properties" and navigate in tabs.

Give your conclusions on the Camera Link Standard

Extension Rings

Mount different extension rings and observe the influence on the acquisition.

Give your conclusions on the use of extension rings

Laboratory Course n° 3
–
Lens & Lighting

Lighting

In order to carry out this step, consult the following documentation:

- Lectures on Machine Vision

Part N°1 – Silhouette Analysis

Setup your system in order to observe the contours of the fork

Propose a solution to measure distances in the images and propose an implementation in order to give the length of the fork.

Part N°2 – Defect Detection on Shiny Surface

Setup your system in order to observe holes & scratches on the surface of the object

For each part, select lens and lighting configuration.

Implement and test your choices

Start grabbing images and ask professor to validate each configuration once everything is OK.

Laboratory Course n° 3
–
Lens & Lighting

Surface Inspection

Machine Vision Configuration

Use LEDs lighting sources to perform the following parts:

1. Center of Oil Filter
2. Magnet
3. Coin detection

For each solution, estimate the budget with the online catalogue of Edmund Optics.

Ask professor to validate once your solution is configured.

Give your conclusions on the lighting techniques.

For each solution estimate the budget with the online catalogue of Edmund Optics:

http://www.edmundoptics.com (choose Poland Country for € cotation)

End of Lab Course

- Disconnect Cables
- Uninstall every lighting devices

Laboratory Course n° 4
–
Compact Vision System

Objective

Study and program a Compact Vision System from KEYENCE

Equipment

- Controller unit (CV-2100)
- Remote control console (OP-42342)
- Monitor cable (RCA - RCA, 2 m)
- Camera (CV-020)
- CV-C3: Camera cable (3 m)
- Monitor CA-MN80
- 24 V DC power supply
- Industrial Parts & Backlight

Software

- Embedded KEYENCE Software

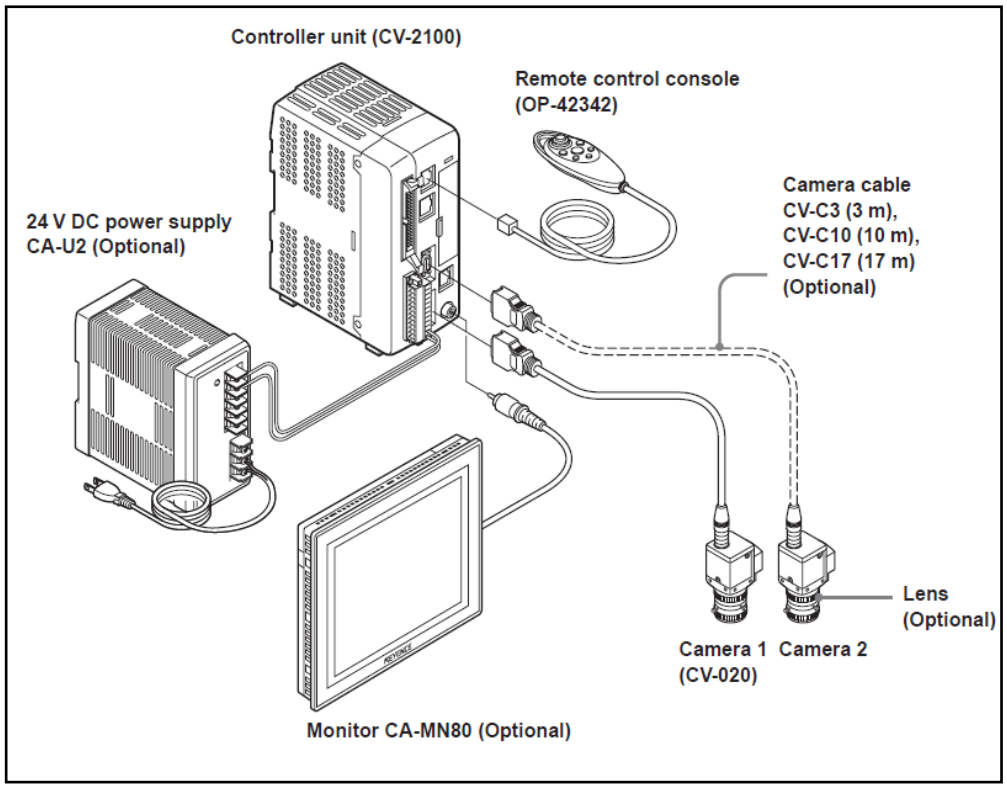
Documentation

- User Manual - High-speed Digital Image Sensor CV-2100

Laboratory Course n° 4
–
Compact Vision System

Installation

In order to carry out this step, consult the user manual



1. Connect Cables

2. Start Grabbing images

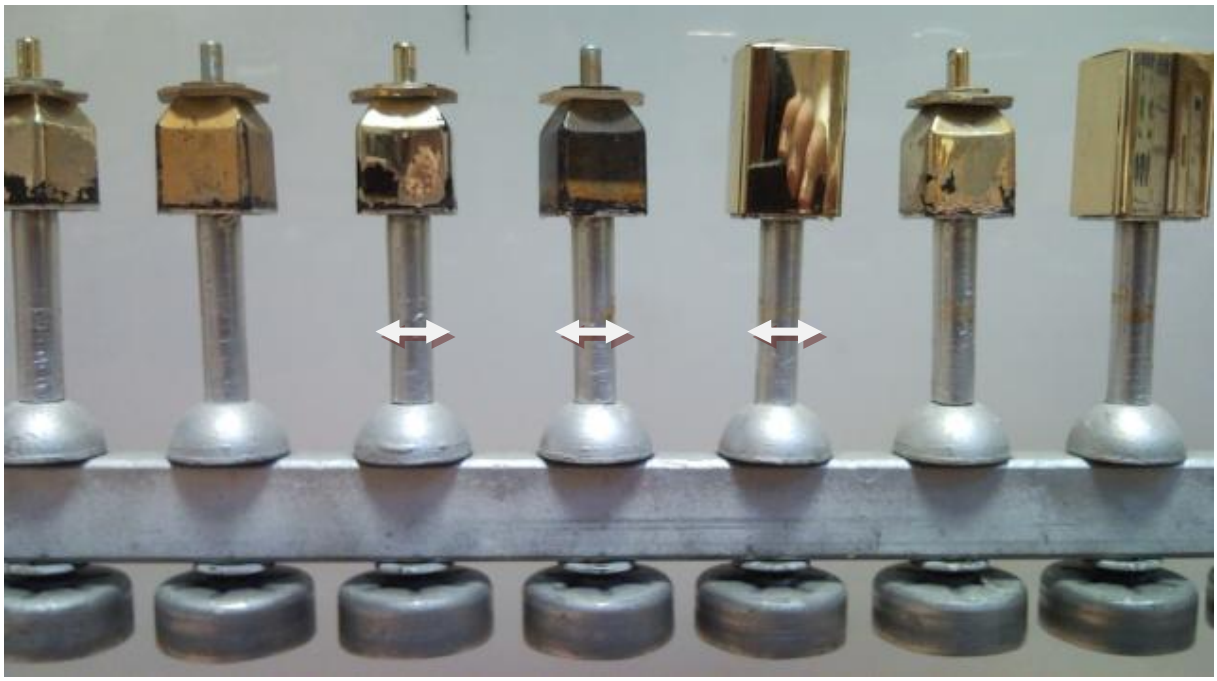
Study user manual chapter 3 for Basic Operation.

Set Camera Settings (chapter 4.2) to obtain a satisfactory image of the shadow of the part.

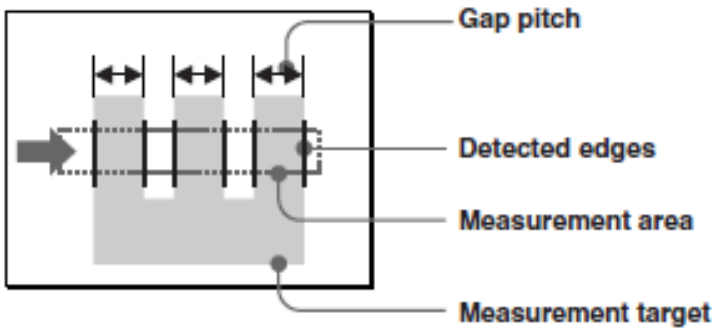
Laboratory Course n° 4
–
Compact Vision System

Edge Detection

You have to program edge detection in order to characterize the holder width on the industrial part.



By using "Edge Pitch" Measurement Mode program the following measure to estimate the width of the holders:

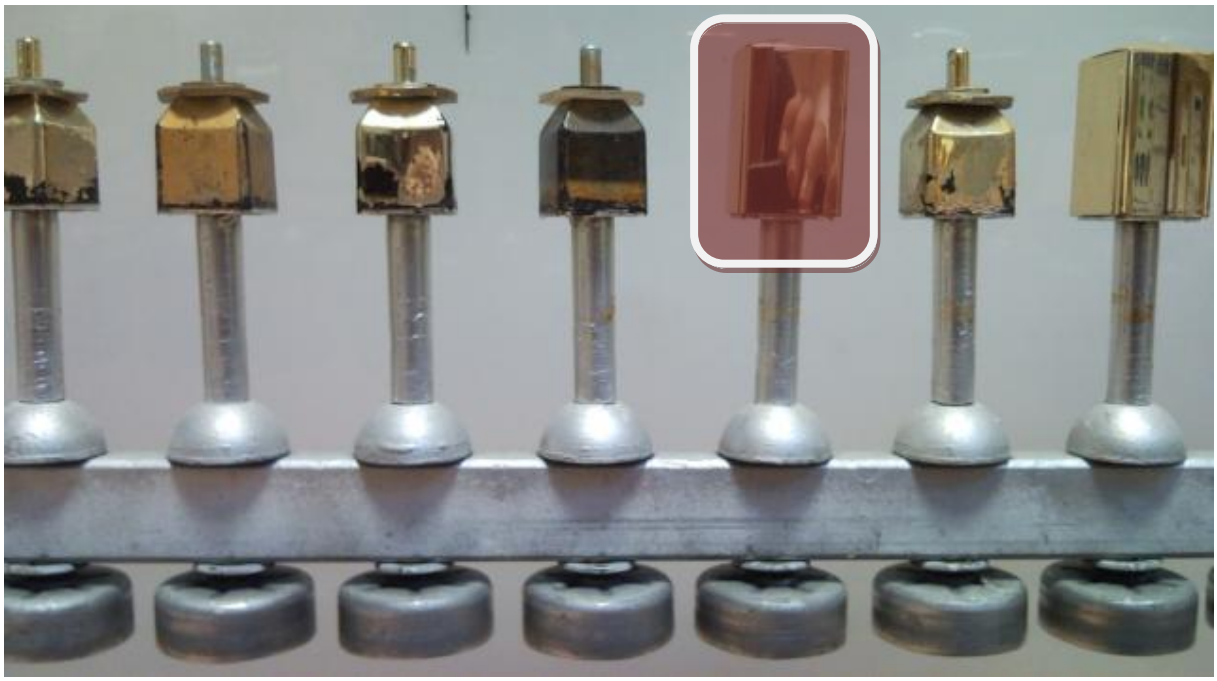


Follow carefully the steps described in "Edge Pitch" chapter (page 4-56 of the user manual).

Laboratory Course n° 4
–
Compact Vision System

Caps Detection

You have to program the detection of perfume caps on the holder.



Select the appropriate tool to perform this detection.

Give your conclusions about compact vision systems

Laboratory Course n° 5
–
Moving Object Imaging

Objective

Study an imaging solution to analyze moving objects. Understand specific features of the system.

Equipment

- PC Computer
- Frame Grabber : DALSA XCELERA-CL LX1
- Digital Camera: DALSA S2-1X-02K40
- 50 mm Lens: PENTAX YF5028A-02
- Video Cables
- 12V Power Supply
- Moving Industrial Parts
- Signal Generator
- Oscilloscope

Software

- CamExpert – TELEDYNE DALSA

Documentation

- User Manual & Brochure for camera & frame grabber
- Lectures on Machine Vision

Laboratory Course n° 5
–
Moving Object Imaging

Camera setting

In order to carry out this step, consult the following documentation:

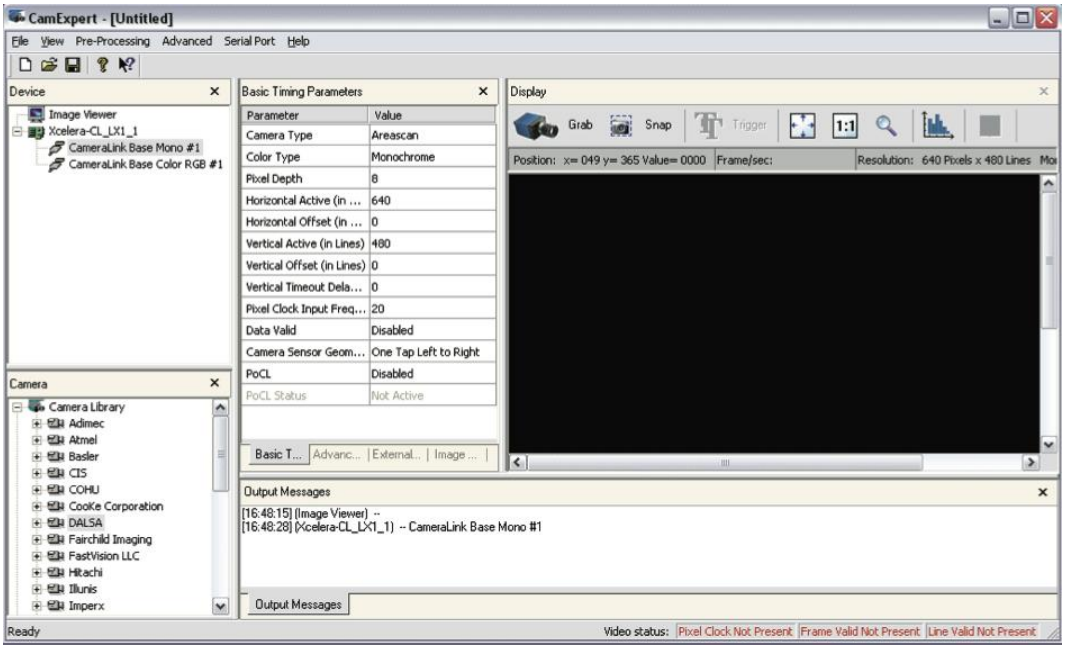
- DALSA Camera & Frame Grabber Brochure
- Lectures on Machine Vision

Switch on the backlight lighting and the moving industrial part motor.

!!! BE CAREFUL WITH MOVING PART !!!

Grab Images

- Start "CamExpert" Software. CamExpert is the camera interfacing tool for frame grabber boards



- Select configuration file for the camera
- Adjust camera settings to obtain a satisfactory image

Start grabbing images and ask professor to validate once everything is OK.

Study camera features: sensor, resolution, video signal …

Study applications of this type of camera

Laboratory Course n° 5
–
Moving Object Imaging

Camera triggering by external signal

In order to carry out this step, consult the following documentation:

- DALSA Camera & Frame Grabber Brochure

!!! ASK PROFESSOR TO VALIDATE YOUR SETUP BEFORE SENDING EXTERNAL SIGNAL TO THE CAMERA !!!

1 - Connect signal generator to the oscilloscope and adjust the external signal provided by the generator in order to trigger the camera.

2 - Connect cables of the camera, frame grabber, signal generator, and oscilloscope in order to trigger the camera by an external signal provided by the generator.

- Start "CamExpert" Software.
- Adjust configuration settings to obtain a triggered image acquisition
- Adjust acquisition frame rate to obtain adapted images of the moving objects. Analyze frame rate modification w/r to the trigger signal frequency

Start grabbing images and ask professor to validate once everything is OK.

Study applications of this type of configuration

Laboratory Course n° 6

3D inspection system

Objectives

3D Inspection with compact vision system

Equipment

• KEYENCE LJ-G 080 System

Software

- Embedded software
- LJ-G Navigator

Documentation

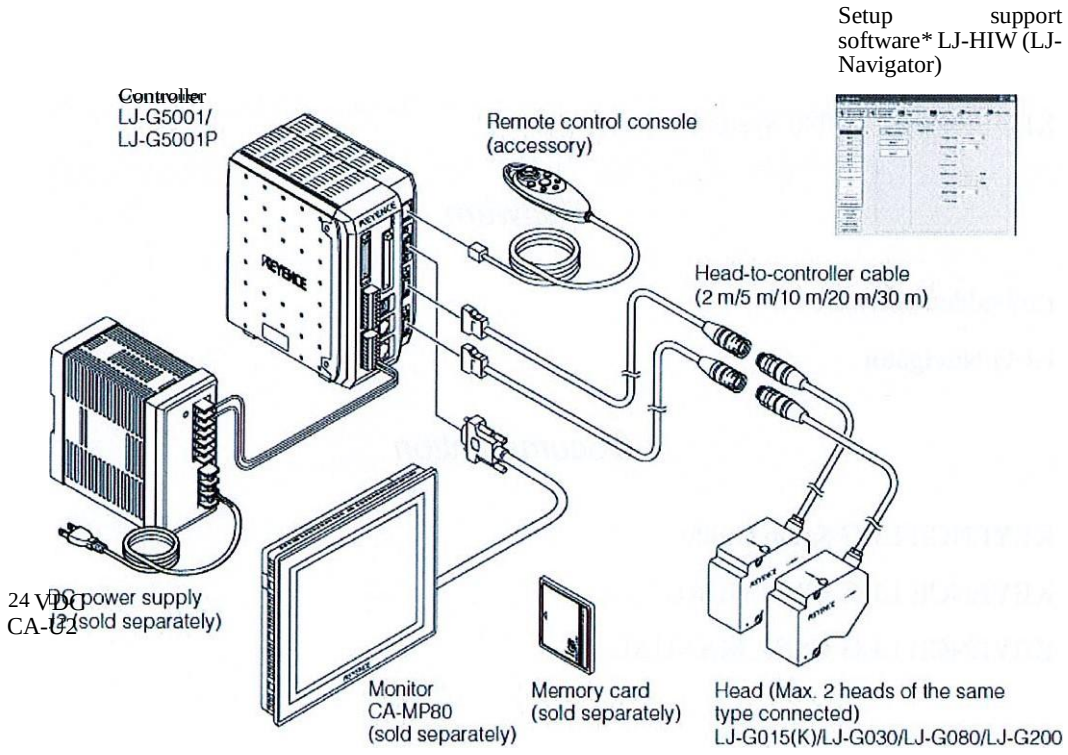
- KEYENCE LJ-G setup Guide
- KEYENCE LJ-G BROCHURE
- KEYENCE LJ-G USER MANUAL

Laboratory Course n° 6

3D inspection system

Setup

Identify the following components;



Read carefully the following pages in Setup Guide: pages 1 to 8

For detailed information read the following pages in User's Manual :
1.1 1-8 1-14 3-2

Laboratory Course n° 6

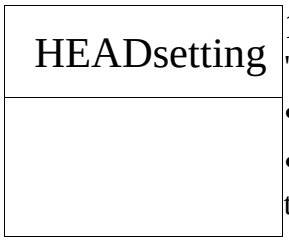
3D inspection system

Programation basics

The workflow for programming is always the same

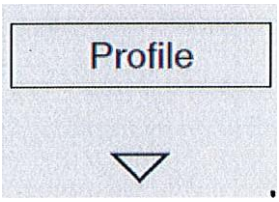
For detailed informations read the following chapter in User's Manual : 5

For each program you will select the following steps

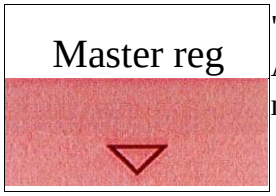


1 -Setting the head sampling conditions
"Head Settings" (Page 5-11)

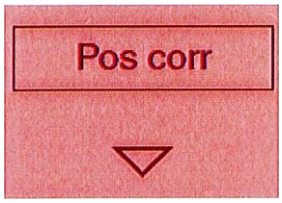
- Set the timing with which the target will be sampled.
- Set the optimal sampling method (sensing method) for the target.



2 - Registering a target to be used as the measurement standard
"Master Registration" (Page 5-48)
Measure the object to be used as the measurement standard (master target). The waveform (profile) from this measurement becomes the standard (registered profile) for measurement mode settings and position adjustments.



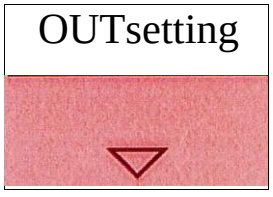
3 - If necessary - Correcting the position and angle of the target
"Position Adjustment" (Page 5-50)
Automatic corrections can be made if the target moves during measurement or is inclined.

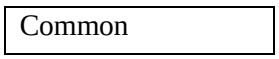


4 - Setting measurement items
"OUT Settings" (Page 5-65)

- Set the measurement type such as "Height measurement. "
- Set the tolerance threshold, average setting, maximum value and other settings.

Profile Settings & "Common Settings" are used in Advanced programming





Laboratory Course n° 6

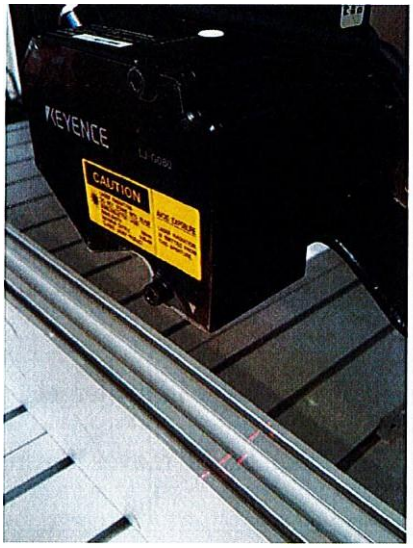
3D inspection system

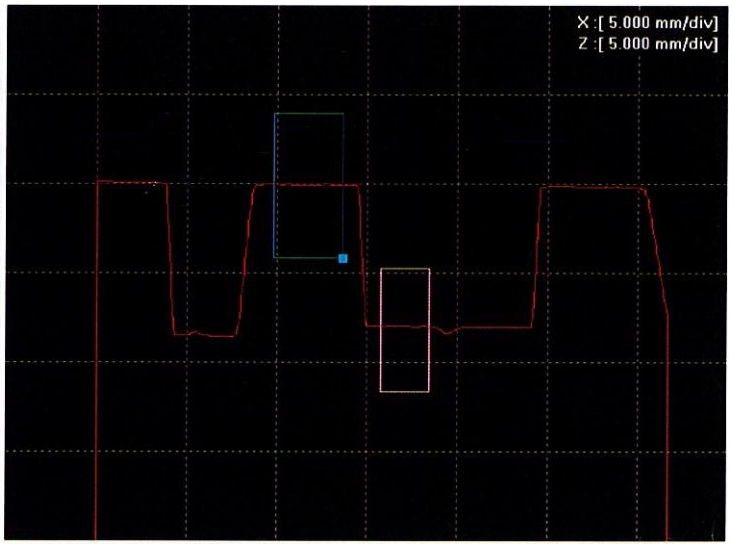
Step Measurement

You have to program step measurement in order to characterize the difference in heigth of the indusffial part.

Measurement

By using "Step" Measurement Mode program the following measure to estimate the height of the profile.





Follow carefully the steps described in "Step" chapter (page 5-85 of the user manual).

Position Adjustment

Enhance the measurement in order to be shift and angle invariant.

Perform shift & angle correction by "Position Adjustment" (Page 5-50)

Laboratory Course n° 6

3D inspection system

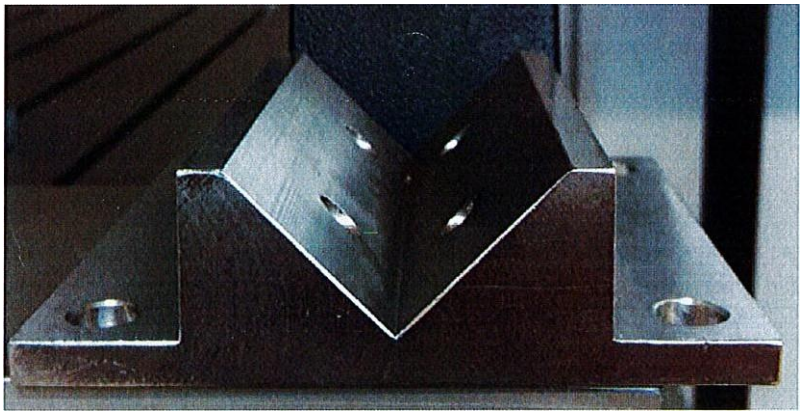
Angle Measurement on Shiny Surface

You have to program angle measurement in order to characterize the angle between two planes on an industrial part.

Measurement

Adjust acquisition settings in order to cope with shiny surface => Menu MFL

Program the system to estimate the angle between the two planes :



Position Adjustment

Enhance the measurement in order to be shift and angle invariant.

Perform shift & angle correction by "Position Adjustment" (Page 5-50)

Holes Detection

Check that you can Detect Holes in the part

Measurement on Cylinders

Realize a program that check the depth difference on the given set of cylinder parts.

One of the object is considered as the reference.

Laboratory course n° 7

Polarization: characteristic curve of LC cell

Introduction

Objective

The main goal is to understand how work a liquid crystal cell and to extract the characteristic curve form an equation obtained with the Stokes model :

Equipment

PC computer

Camera via USB cable

Meadowlark switchable polarization Liquid Crystal modulator + Meadowlark LC Driver (the CD device is attached to the controller output 1)

Two linear polarizers

Mounting posts and post holders

Lighting Device and monochromator

Power source of the photodiode ; NI 6009 I/O link to the computer

Software

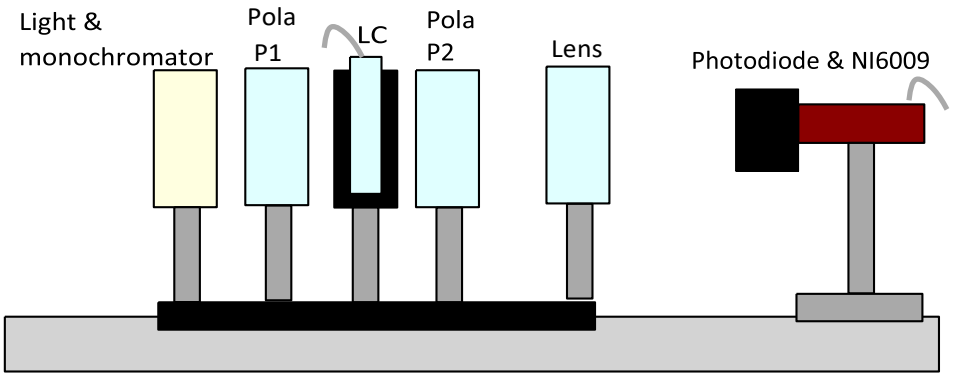
Meadowlark LC Software

Matlab

To prepare the lab

Getting started

Identify the different parts of the setup :



For the experiment, the angle of the first polarizer is 0° (horizontal) and the position of the fast axis of the LC cell is 45°.

Consider the Mueller matrix of the two polarizers and the one of the LC cell assumed to be an ideal retarder (see slide xx of the course). The intensity of the light source is noted I0.

1) Express the Stokes vector of the light just after the polarizer P1.
2) Express now the Stokes vector after the retarder (LC cell) and it's expression when the angle of the retarder is 45°.
3) Express the Stokes vector of the light emerging from the polarizer P2 oriented at an angle α .
4) Deduce the intensity on the photodiode and express it when α is 0° or 90° and from the two expressions extract the expression of cos (ϕ)with ϕ the phase modulation angle of the retarder.

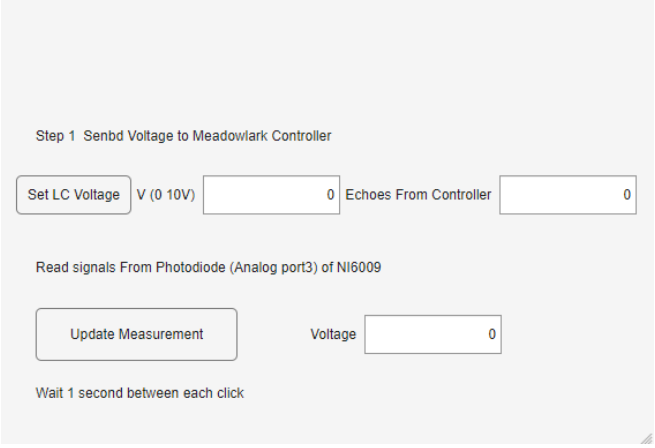
Manipulations

Switch on the light source, and check that after the monochromator the selected wavelength is 633 nm.

Swith on the computer, the Meadowlark controller.

Notice that the photodiode module is plugged to a power source. Switch on this power source and tune it to have a maximum range of 10V and check that it is plugged in order to have an inverse polarity

Launch Malab and the acquisition application "CamilLCVR.mlapp" into the folder of the windows board. This open a graphical interface that permits to set a voltage to the Liquid Cristal cell and then read a voltage given by the photodiode.



Manually test several voltages settings and observe the effect on the light reaching the photodiode by using a piece of paper.

First series of measurements

With the LC Cell control panel you will tune the voltage from 0 to 10 Volt by steps of 0.1 V

For each step :

1. Adjust the output polarizer P2 to 0° and take the voltage issued from the photodiode,

2. The adjust the polarizer P2 to 90° and take the voltage issued from the photodiode

Change the voltage of the LC cell and repeat steps 1 and 2

Now you are able to evaluate the angle ϕ and graph the curve: phase shift (ϕ) depending of the tuning voltage.

Second series

You can repeat the same measurement by changing the wavelength on the monochromator, let's say 532 nm.

Comment the differences on the characteristic curve